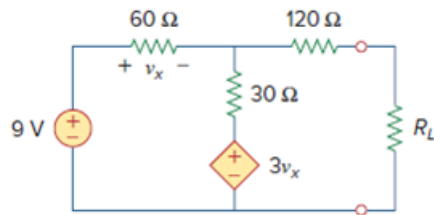


Problem

Determine the value of R_L that will draw the maximum power from the rest of the circuit in Fig. 4.52. Calculate the maximum power.

Figure. 4.52:

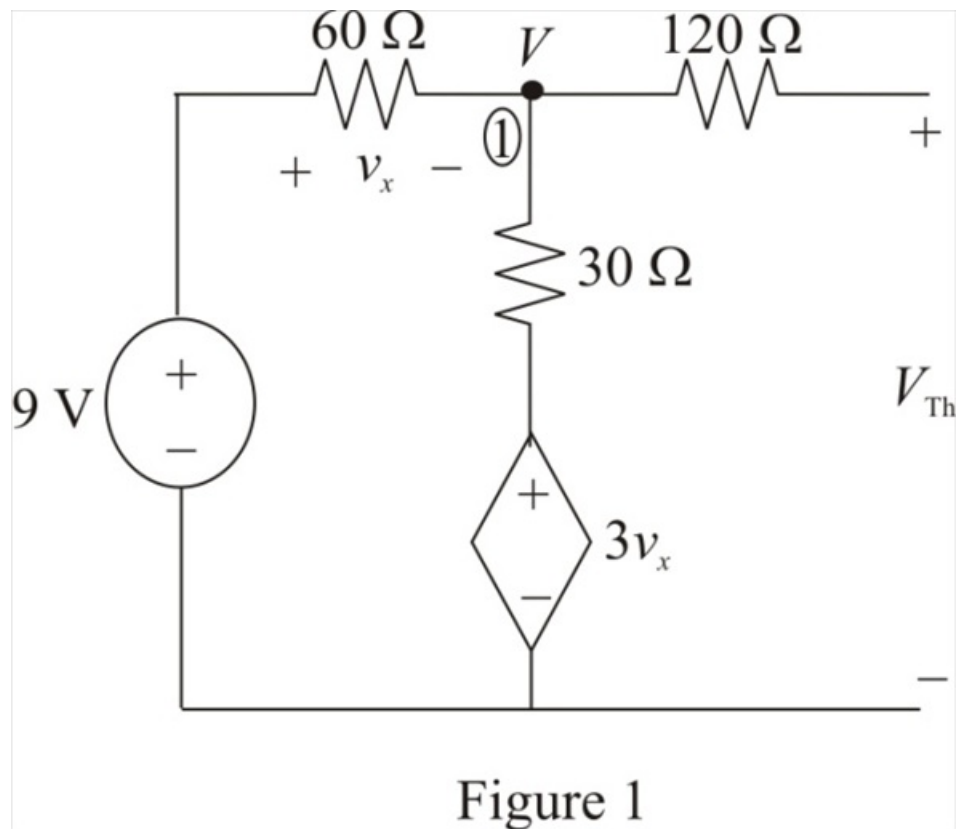


Step-by-step solution

Step 1 of 8

Refer to Figure 4.52 for the circuit in the text book.

Disconnect the load resistance AND calculate the open circuit voltage shown in Figure 1.



Step 2 of 8

Consider the expression for the voltage, v_x .

$$v_x = 9 - V$$

Therefore, the expression for the voltage v_x is $9 - V$.

Apply the Kirchhoff's current Law at node 1.

$$\frac{V - 9}{60} + \frac{V - 3v_x}{30} = 0$$

$$\frac{V - 9 + 2V - 6v_x}{60} = 0$$

$$3V - 9 - 6v_x = 0$$

Step 3 of 8

Substitute $9 - V$ for v_x in the equation.

$$3V - 9 - 6(9 - V) = 0$$

$$3V - 9 - 54 + 6V = 0$$

$$9V - 63 = 0$$

$$9V = 63$$

$$V = \frac{63}{9}$$

$$= 7 \text{ V}$$

Here,

V represents the Thevenin's voltage V_{Th} .

Determine the value of Thevenin's voltage, V_{Th} .

$$V_{Th} = 7 \text{ V}$$

Therefore, the value of the Thevenin's voltage V_{Th} is 7 V .

Step 4 of 8

Short the output terminals and calculate the short circuit current I_{sc} shown in Figure 2.

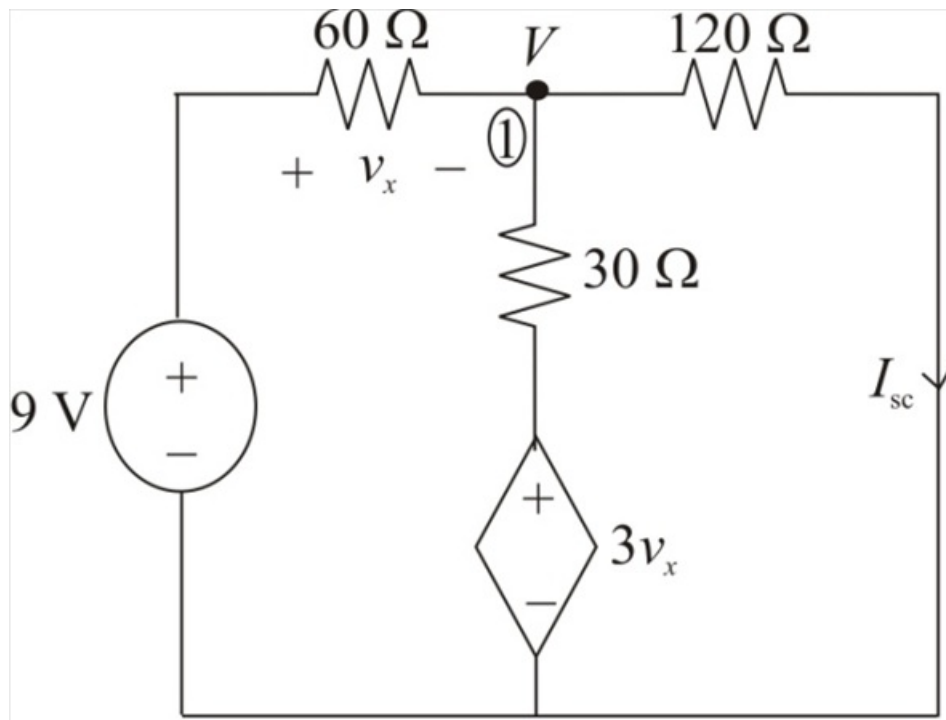


Figure 2

Step 5 of 8

Apply the Kirchhoff's current Law at node 1.

$$\frac{V-9}{60} + \frac{V-3v_x}{30} + \frac{V}{120} = 0$$

$$\frac{2V-18+4V-12v_x+V}{120} = 0$$

$$7V-18-12v_x = 0$$

Substitute $9-V$ for v_x in the equation.

$$7V-18-12(9-V) = 0$$

$$7V-18-108+12V = 0$$

$$19V-126 = 0$$

$$19V = 126$$

$$V = \frac{126}{19} \text{ V}$$

Therefore, the value of the voltage V is $\frac{126}{19} \text{ V}$.

Step 6 of 8

Calculate the value of the short circuit current, I_{sc} .

$$I_{sc} = \frac{V}{120}$$

Substitute $\frac{126}{19}$ V for V in the equation.

$$\begin{aligned} I_{sc} &= \left(\frac{1}{120} \right) \left(\frac{126}{19} \right) \\ &= 55.26 \text{ mA} \end{aligned}$$

Therefore, the value of the short circuit current I_{sc} is 55.26 mA.

Step 7 of 8

Calculate the value of the Thevenin's resistance, R_{Th} .

$$R_{Th} = \left| \frac{V_{Th}}{I_{sc}} \right|$$

Substitute 55.26 mA for I_{sc} and 7 V for V_{Th} in the equation.

$$\begin{aligned} R_{Th} &= \frac{7}{55.26 \text{ m}} \\ &= 126.67 \Omega \end{aligned}$$

Therefore, the value of the Thevenin's resistance R_{Th} is 126.67 Ω .

For the load to absorb the maximum power, its value should be equal to the Thevenin's resistance.

$$\begin{aligned} R_L &= R_{Th} \\ &= 126.67 \Omega \end{aligned}$$

Therefore, the required value of the load resistance R_L is 126.67 Ω .

Step 8 of 8

Calculate the maximum value of the power p_{max} absorbed by the load.

$$p_{max} = \frac{(V_{Th})^2}{4R_{Th}}$$

Substitute 126.67 Ω for R_{Th} and 7 V for V_{Th} in the equation.

$$\begin{aligned} p_{max} &= \frac{(7)^2}{4(126.67)} \\ &= \frac{49}{506.68} \\ &= 96.71 \text{ mW} \end{aligned}$$

Therefore, the maximum value of the power p_{max} absorbed by the load is 96.71 mW.